Low-Cost Manure Nutrient Content Measurement Kit

A Portable Tool for On-Farm Analysis of Nutrient Content in Organic Manure

Project Title
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Abstract

This project introduces a low-cost, portable nutrient measurement kit designed to help users assess the nutrient content of Farmyard Manure (FYM) and Liquid Organic Manure (LOM). By analyzing key nutrients like nitrogen (N), phosphorus (P), and potassium (K), along with pH, moisture, and conductivity levels, users can make informed decisions to optimize crop nutrition. No more guesswork or expensive lab tests - just quick, reliable results right on your farm. Our goal? To help farmers make smarter decisions about fertilizing, save money, and keep their soil happy and healthy.

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Introduction

Nutrient management is crucial for optimal crop growth and soil health. Many farmers, however, lack access to affordable tools for testing nutrient content in organic manure. This kit provides an accessible solution, enabling farmers to analyze essential nutrients on-site without needing expensive lab equipment. By offering quick and accurate feedback on manure quality, the kit helps to enhance productivity and sustainability in agricultural practices.

Project Objectives

- Develop a low-cost, portable tool to measure N, P, K, pH, moisture, and conductivity in organic manure.
- Provide an easy-to-use device that delivers reliable, actionable results for small-scale farmers.

Problem Statement

At the village level, assessing the nutrient content of Fermented/Liquid Organic Manure (F/L OM) is challenging for users, leading to inconsistent quality and inefficient use. For governments purchasing manure, the lack of affordable and simple testing tools makes quality control difficult, affecting crop yield and sustainability.

There’s often a struggle to understand the nutrient composition of organic manure, which leads to improper application and potential nutrient imbalances in soil. Existing commercial testing equipment is often too expensive.

Literature review

Numerous nutrient testing methods exist, but high costs and complexity limit their practicality for small-scale users. Simple colorimetric techniques and sensors like TCS34725 (for color detection), pH sensors show promise in reducing costs while maintaining accuracy. This project builds on such techniques, providing a comprehensive nutrient measurement system with key improvements for affordability and ease of use.

The State Government of Chhattisgarh, India, started buying cow dung under the Godhan Nyay Yojana scheme from July 2020. This initiative aims to promote sustainable agriculture and waste management practices.

According to the Chhattisgarh government's cabinet decision on July 15, 2020, the procurement price of cow dung is fixed at ₹2 per kilogram.This price is applicable for cow dung collected from livestock owners in villages with cowsheds, and the selling price of vermicompost produced under the scheme is capped at ₹8 per kilogram for farmers through cooperative societies.

System Design

- Hardware Overview

  ESP32

  - The ESP32 is a low-cost, low-power System on Chip (SoC) microcontroller developed by Espressif Systems.With built in WiFi and Bluetooth.The ESP32 is widely used in various IoT applications, including wireless communication, robotics, home automation, and audio/video processing.

  pH sensor with amplifier circuit

  - A pH sensor is a device used to measure the acidity or basicity of a solution, typically expressed as a pH value between 0 and 14. pH

sensors are essential in various fields, including chemistry, biology, environmental science, and industrial processes.

TCS34725 (RGB sensor)

- The TCS34725 is a digital color light-to-digital converter (CLD), featuring a high-precision RGB and ambient light sensor with an IR blocking filter. This sensor measures the intensity of red, green, and blue (RGB) light, as well as clear light, and provides a digital output.

OLED

- The SSD1306 is a single-chip CMOS OLED/PLED driver with a controller for organic/polymer light-emitting diode (OLED) dot-matrix graphic display systems. It consists of 128 segments and 64 commons, designed for Common Cathode type OLED panels.

Push button

- A push-button is a simple switch mechanism used to control some aspect of a machine or process.
  Capacitive soil moisture sensor.

Jumper wires

- Connecting wires.

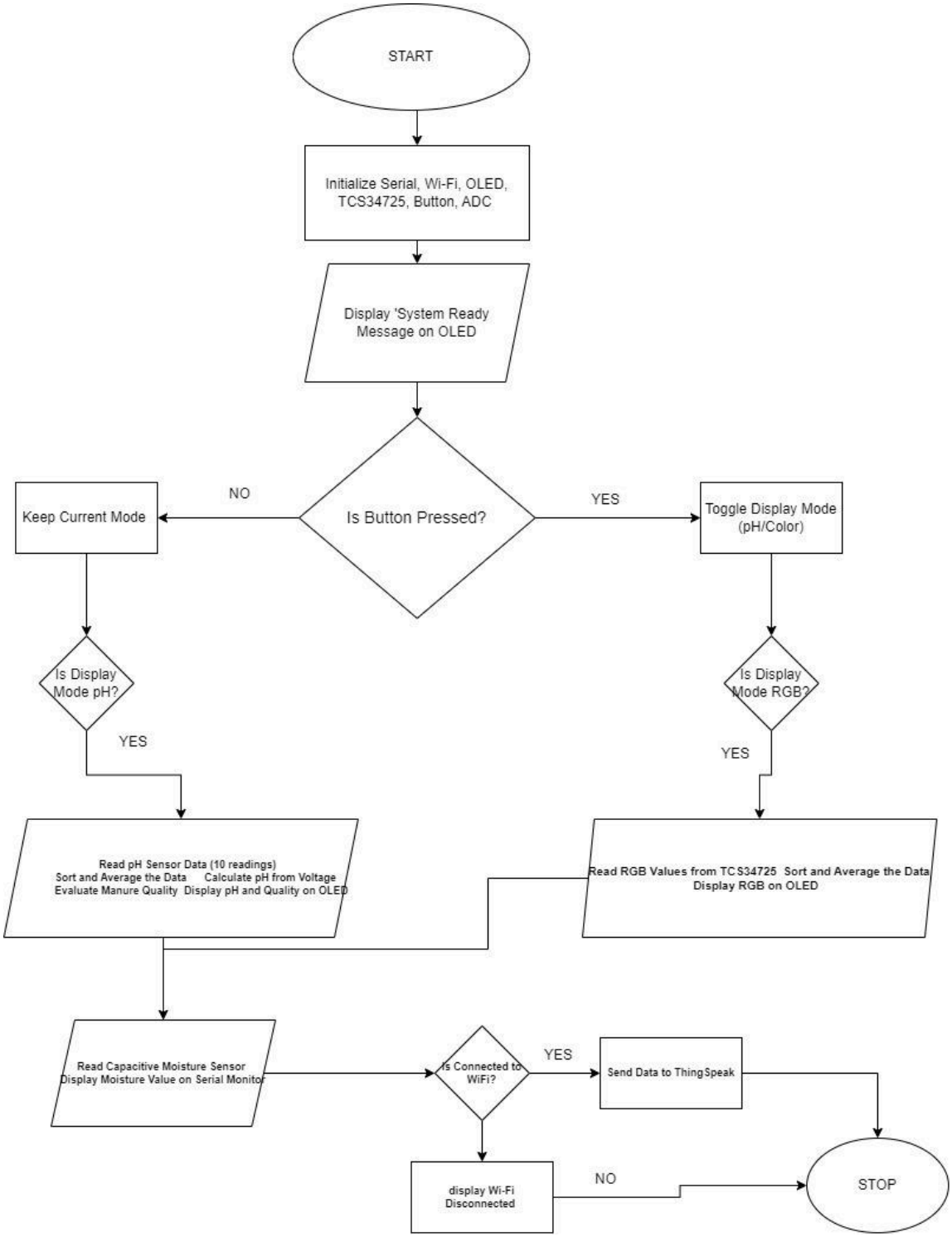
- ## Software Overview

This code connects an ESP32 to Wi-Fi and gathers data from a pH sensor, capacitive soil moisture sensor, and an RGB color sensor. The pH and moisture values are displayed on an OLED screen, and a button toggles the display between pH and color modes. The code calculates the pH based on sensor readings and displays RGB values from the color sensor. It also sends the pH, moisture, and RGB data to ThingSpeak for remote monitoring. This code ensures regular data updates to ThingSpeak while displaying real-time sensor information on the OLED.

We have used Arduino IDE to code , compile and upload our sketches using c and c++ programming language.Libraries used :-

- **Wire.h** - for I2C communication protocol.
- **U8g2lib.h** - for displaying info in SSD1306 OLED as the SSD1306 library was not working for our OLED.
- **Adafruit_TCS34725.h** - for using the RGB sensor
- **Adafruit GFX Library** - This is the core graphics library for all our displays, providing a common set of graphics primitives (points, lines, circles, etc.). It needs to be paired with a hardware-specific library for each display device we carry (to handle the lower-level functions).
- **HTTPClient.h -** HTTPClient.h is a C++ header file that provides a library for interacting with web servers from an Arduino-based microcontroller. It's specifically designed for use with the Arduino platform and allows developers to send HTTP requests.

FLOW CHART

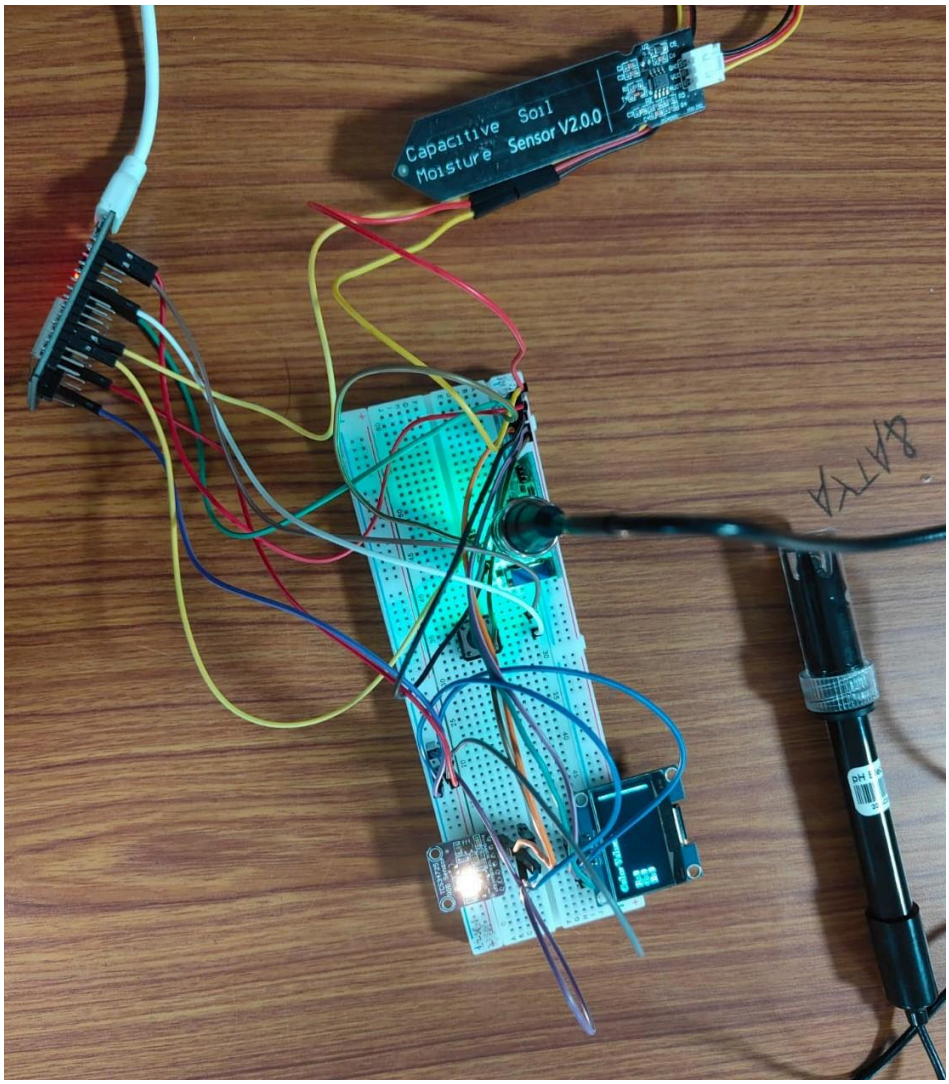

START
Initialize Serial, Wi-Fi, OLED, TCS34725, Button, ADC
Display 'System Ready Message on OLED
Is Button Pressed?
NO
YES
Keep Current Mode
Toggle Display Mode (pH/Color)
Is Display Mode pH?
Is Display Mode RGB?
YES
YES
Read pH Sensor Data (10 readings)
Sort and Average the Data Calculate pH from Voltage
Evaluate Manure Quality Display pH and Quality on OLED
Read RGB Values from TCS34725 Sort and Average the Data
Display RGB on OLED
Read Capacitive Moisture Sensor
Display Moisture Value on Serial Monitor
Is Connected to WiFi?
YES
Send Data to ThingSpeak
display Wi-Fi Disconnected
NO
STOP

Methodology

- Development Process
    - **Assembling Hardware** : Connect color sensor, pH sensor, moisture sensor, and microcontroller to create the basic setup.
    - **Sensor Calibration**: Calibrate each sensor (pH, color, moisture) using standard solutions to ensure accuracy.
- Algorithm Explanation
- Testing and Calibration

Implementation

Hardware assembly:



Software Code Structure:

1. **Libraries**:
    - `Wire.h`, `U8g2lib.h`: To handle I2C communication with the OLED display.
    - `Adafruit_TCS34725.h`: To interface with the TCS34725 color sensor.
    - `WiFi.h`, `HTTPClient.h`: To connect to Wi-Fi and send HTTP requests to ThingSpeak.
2. **Wi-Fi Setup**:
    - The ESP32 connects to Wi-Fi using the credentials provided and checks the connection status in the `setup()` function.
3. **ThingSpeak Configuration**:
    - `sendDataToThingSpeak()` function sends the sensor data (pH, moisture, and color values) to ThingSpeak via an HTTP GET request. This allows remote monitoring and visualization of the data on the ThingSpeak platform.
4. **pH Sensor Handling**:
    - Reads analog values from the pH sensor and converts them to a pH value using a calibration factor. The code averages multiple readings to improve accuracy.
    - Based on the pH value, the system evaluates manure quality using predefined ranges and displays this on the OLED screen.
5. **Color Sensor Handling**:
    - Captures RGB data using the TCS34725 sensor. These values represent color changes (useful for nutrient detection based on chemical reactions).
6. **Moisture Sensor**:
    - Reads the analog value from a capacitive soil moisture sensor and prints it to the Serial Monitor.
7. **Button Handling**:
    - A button is used to switch between displaying pH values or color values on the OLED screen.

Experimental Results

- Testing Scenarios

  Testing for the pH sensor was done, which was giving an almost stable reading with accuracy of 85% when it was dipped in tap water which was slightly alkaline in nature.

  Testing of Color sensor TCS43725 was done with reference to Rubik's cube. It was correctly sensing R, G, B but was unable to distinguish properly between orange and yellow.

- Comparison to Expected Outcomes

  Results showed reliable accuracy in measuring pH and color, though minor adjustments were needed for calibration as there were fluctuations in the readings.

Analysis and Discussion

All of the components are working properly and giving results with certain accuracy, but time-to-time calibration is required for pH and color sensor so that we could get maximum accuracy as there is still an error rate of 10-15%.

Challenges and Limitations

Challenges

- **Technical**: Sensitivity of color sensors in varying light.
- **Logistical :** we have reused a broken OLED panel in which one strip was not working by changing the orientation.
- **Calibration and Maintenance**: Regular calibration is essential for sensors like pH and color sensors to maintain accuracy.

Limitations

- **Sensor Accuracy**: Limited by sensor quality, which could affect readings in high-moisture or low-light settings.
- **Sensors :** Careful handling the pH sensor is required as it is highly sensitive

Future Work and Enhancements

1. **IoT Integration**: Adding Bluetooth or Wi-Fi for mobile data logging.
2. **Enhanced Sensors**: Use of more sensitive sensors for greater accuracy.
3. A proper printed board with all the components printed on it for easily handling of the product.

Conclusion

Our Farm-Friendly Manure Nutrient Analyzer might not be the fanciest kit on the block, but it's a practical, affordable tool that puts real power in farmers' hands. By making nutrient testing accessible to everyone, we're hoping to promote smarter fertilizer use, healthier soils, and more sustainable farming practices. With future improvements, this kit could serve as an essential tool for precision agriculture, fostering better crop yield and soil health.

References

1. Arduino forums
2. Stackoverflow
3. ESPio tutorials
4. Adafruit examples
5. GitHub
6. Datasheets -
    - ESP 32
    - TCS34725
    - Capacitive Soil Moisture sensor
    - DHT11
    - OLED
    - pH sensor